Eastern Bering Sea Shelf Bottom Trawl Survey
Gear Description for 83-112 Eastern Trawl

Netting: Body and wing 4" stretched measure (including length of one knot) 60 T nylon, three-strand twist, preshrunk and dyed green.
Intermediate and codend - 3½" stretched measure 96 T nylon, three-strand, twist, preshrunk and dyed green.
Codend liner - 1¼" stretched measure 18 T nylon, three-strand twist, preshrunk and dyed green.

Headrope: 83' 9" of ½" (6×19) galvanized fiber core wire rope wrapped with ⅜" polypropylene rope.
Both eyes have ½" gusseted thimbles.
The headrope doesn't include the length of either eye.
Length is measured from the top of the nicro sleeve to the top of nicro at the other end.
Top wings are hung over 36'.

Footrope: 111' 9" plus thimbled eyes of ⅝", (6×19) galvanized, fiber core wire rope, wrapped with ½" polypropylene rope.
Loops of 5/16" galvanized, proof coil chain, (approx. 170') are tied to wrapped footrope by passing two fathoms of double 21 T nylon twine through every tenth link starting at the top of the nicro forming the eye splice.
Chain links are secured to footrope every 8" throughout length of wing forming 76 chain hangings over 50' each wing.
Busom has 16 chain hangings equally spaced over 11' 9".
There are four bars (2 meshes) per hanging in wings and 4 meshes per hanging in busom, hung with 96 T nylon round braid hanging twine.
After the net is hung, split pieces of heavy rubber hose are served around footrope, passing the hose twice between each chain hanging for the length of the footrope.

Breastlines: ½" (6×19) galvanized wire rope wrapped with ⅜" polypropylene rope.
Top and bottom breastlines are 8', measured from the top of the nicro press forming one eye through the bearing point of the other eye.

Riblines: ¾" Samson 2 and 1® Duralon braided trawl rope.
Riblines are hung at 97% of stretched measure of the gored seam.
All measurements are made with 400 lbs of tension on rope.
Gored seams are attached to riblines using white, untreated 60 T braided nylon hanging twine used to tie a benzel every 16".

Flotation: Seventeen 8" aluminum side lug floats along each wing and seven 8" floats in center.
Total of 41 floats along headrope spaced 24¼" apart.
Buoyancy of 6.3 lbs each (258 lbs total).

Side seams: Side seams are laced in top and bottom individually gathering 3 meshes (4 knots) using white, double 21 T perma-grip® (or like kind) nylon three-strand twine.
Top and bottom panels are then laced together using green, double 21 T perma-grip® (or like kind) nylon three-strand twine.

Codend: 3½" stretched measure (including one knot) nylon, dyed green, 96 T nylon three-strand mesh.
Codend is a "double wall" construction.
Four panels of 96 T web cut 64 meshes long by 120 meshes deep.
Two meshes on each side are laced together, leaving 60 "open meshes" per panel.
Gored seams are laced together using 60 T round braid hanging twine and hung to riblines of ¾" Samson 2in1® Duralon braided trawl rope.
Riblines in codend are hung at 90% of stretched measurement of gored seams.
Riblines are measured under 400 lbs of tension.
Codend is closed at aft end using 24-2½" x 5/16" galvanized steel rings.
A ¼" Duralon braided rope is passed through 5 selvage meshes and a ring is attached to the rope using a cow hitch every 12" leaving five open meshes between each ring.
The bag is then closed using a ⅝" - ¾" hauling clip.
A liner of 1¼" three strand twine nylon, 360 meshes long by 200 meshes deep is hung on the inside of the bag 78 meshes up from terminal end.

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Materials List for 83-112 Eastern Trawl

Specifications of wire, chain, and rope include rated breaking strength (BS), specific gravity (SG), and density.

WEB: Mesh sizes listed are given in "stretched measure", a standard method of measuring mesh size that includes the length of one knot.
Body and wings ~ 4" # 60T three strand nylon - dyed green.
Intermediate ~ 3½" # 96T three strand nylon, dyed green, single wall construction.
Codend ~ 3½" # 96T three strand nylon, dyed green, double wall construction.
Liner ~ 1¼" # 18T three strand nylon - dyed green.
Chaffing gear ~ 6", 6-mm polyethylene knotted web.

WIRE ROPE: ½" 6×19 galvanized fiber core (BS 9,700 kg, 0.62 kg/m), eyes are formed using galvanized reinforced thimbles.

CHAIN: 5/16" galvanized proof coil (BS 4,750 kg, 1.21 kg/m).

FLOATS: 8" side lug trawl floats, depth rated to a minimum 400 fm, 258 lbs of total buoyancy.

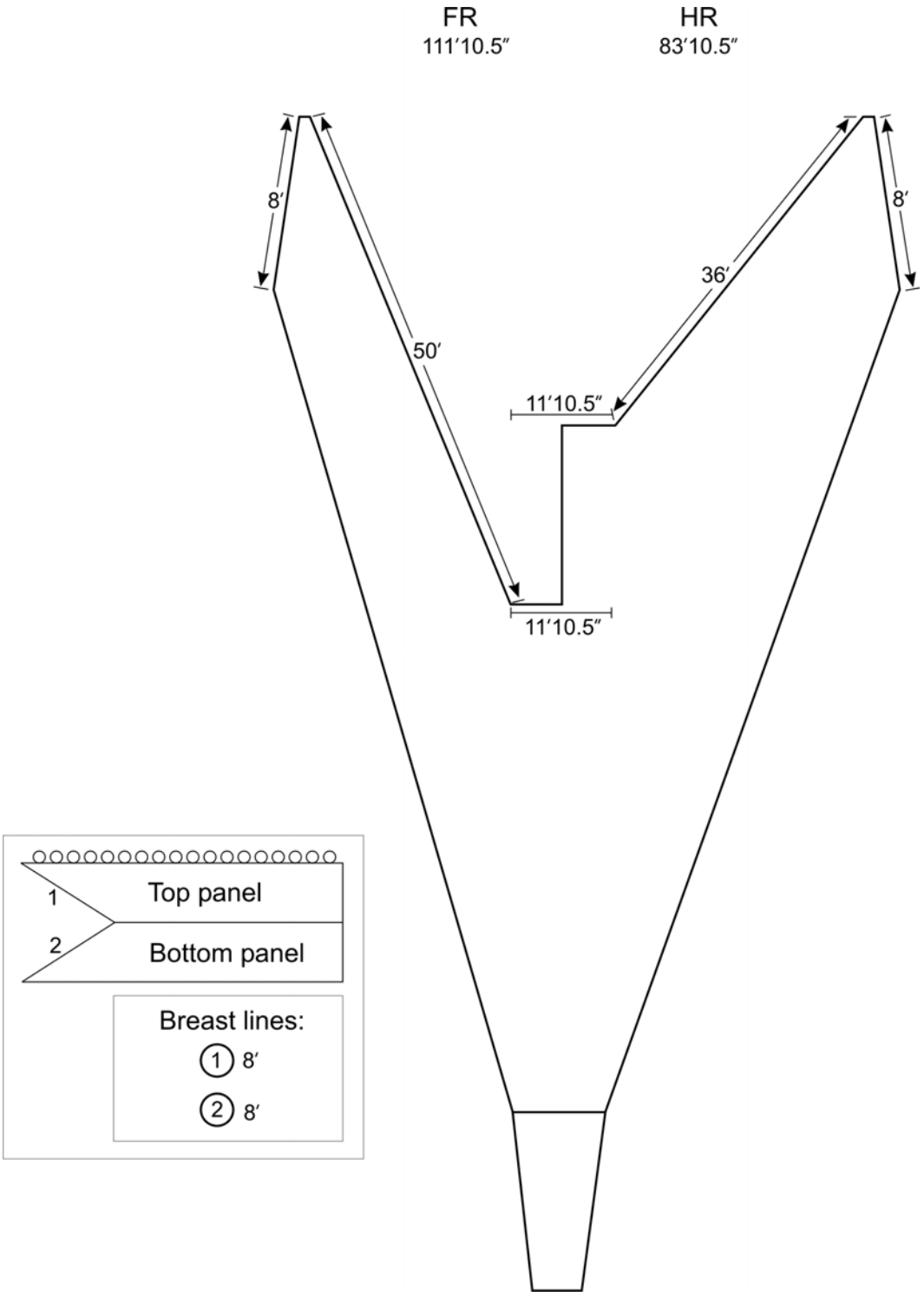
RIBLINES: ¾" Samson Duralon™ 2-in-1® stable braid, white, light bonding (BS 8,800 kg, SG 1.83, 0.27 kg/m).

TWINE: Sewing panels together: # 60T three strand nylon.
Lacing seams: double, white # 21T three strand nylon.
Joining seams together: double, green # 21T three strand nylon.
Hanging on framing lines: white, # 96T round braid hanging twine.
Fly mesh at end of bag- # 182T round braid hanging twine

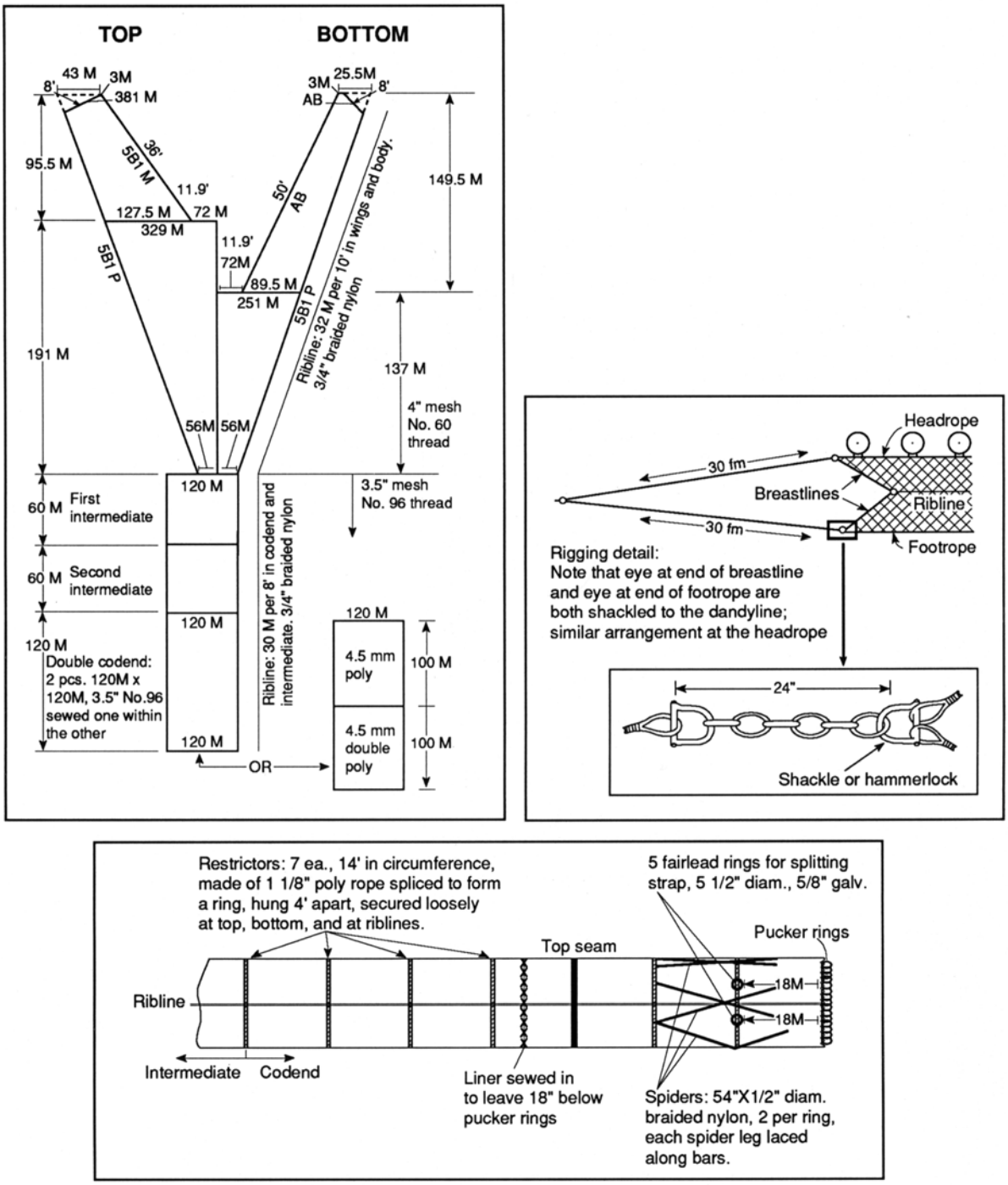
ROPE: Float lines - 1/4" polypropylene (BS 570 kg, SG 0.95).
Pucker ring lines- Samson 1/4" Duralon™ 2-in-1®(BS 1,270 kg, SG 1.38).
Restrictor rope- 1 1/8" Poly Plus™ three strand (BS 10,740 kg, SG 0.91).
Splitting strap- 3/4" Spectra™ braided rope with eye each end (BS 21,773 kg, SG 0.98).
Splitting ring lines- ½" Samson Duralon™ 2-in-1® stable braid (BS 4,477 kg, SG 1.38).
Hole through trawl float line- ½" hollow braid Poly-Plus™ (BS 2,721 kg, SG 0.91).

FOOTROPE: ⅝" 6×19 galvanized, fiber core wire rope with reinforced thimbled eyes.
½" three strand polypropylene rope served around wire rope.
5/16" galvanized proof coil chain.
Split heavy rubber hose served between chain hangings.
Footrope setback/extension, ½" long link alloy chain connected to footrope with a ⅝" hammer lock. Total length including connecting hardware - 24".
Splitting rings - 5½" × ¾".
Pucker rings at end of codend- 2½" × ¼".

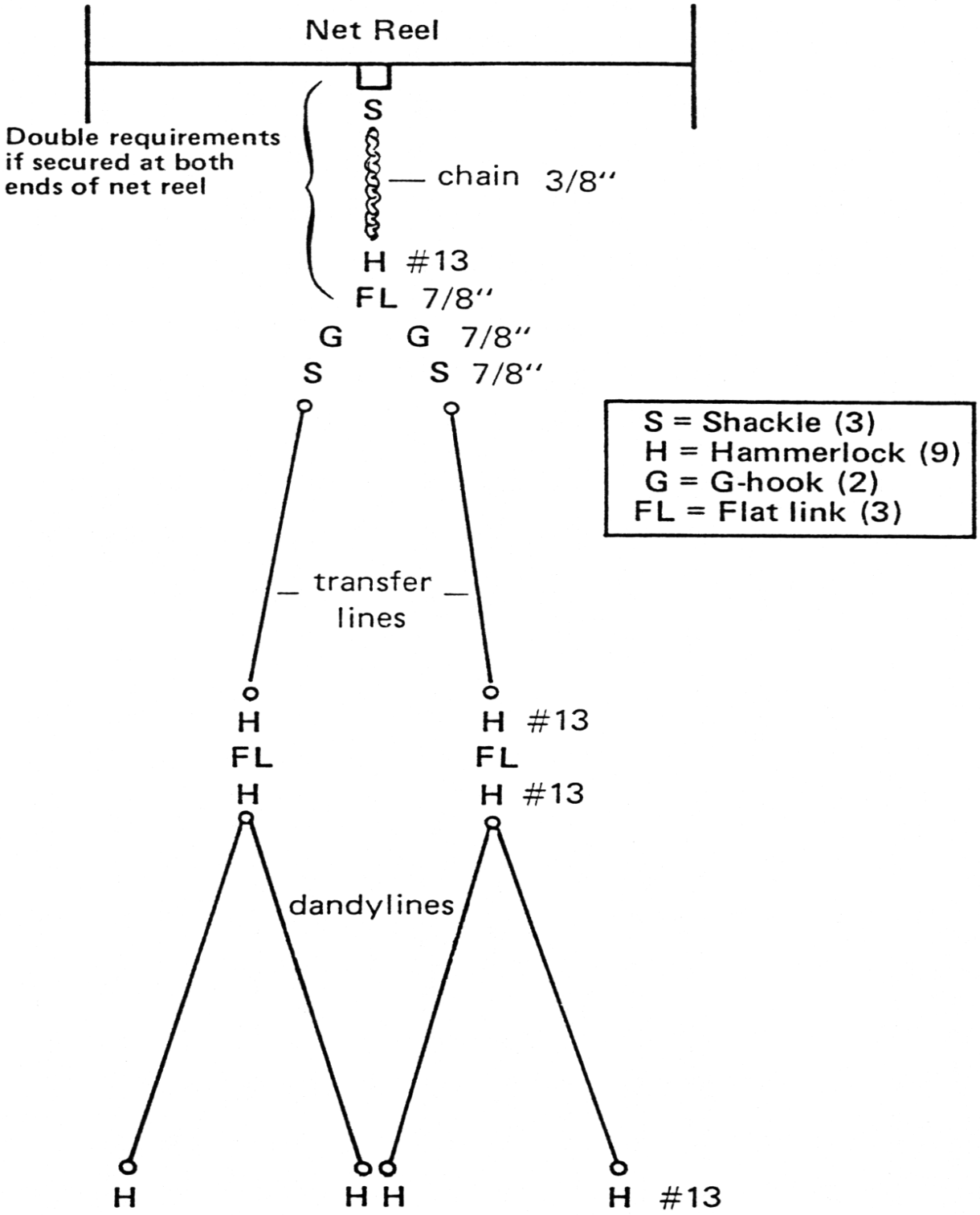
Eastern Bering Sea Shelf Bottom Trawl Survey
Framing Lines for 83-112 Eastern Trawl



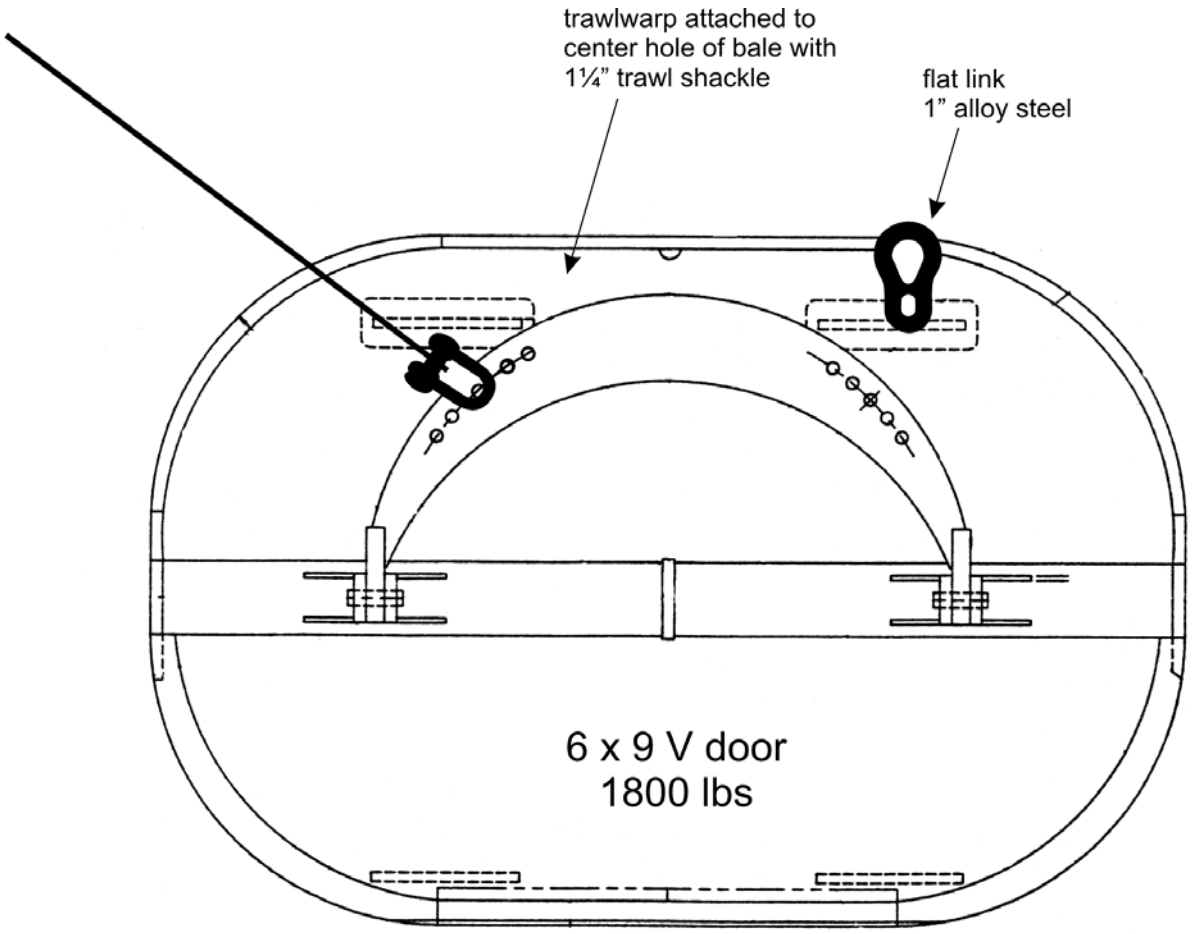
Eastern Bering Sea Shelf Bottom Trawl Survey
Net Plan for 83-112 Eastern Trawl
(“Cut Plan” = Total Mesh Counts)



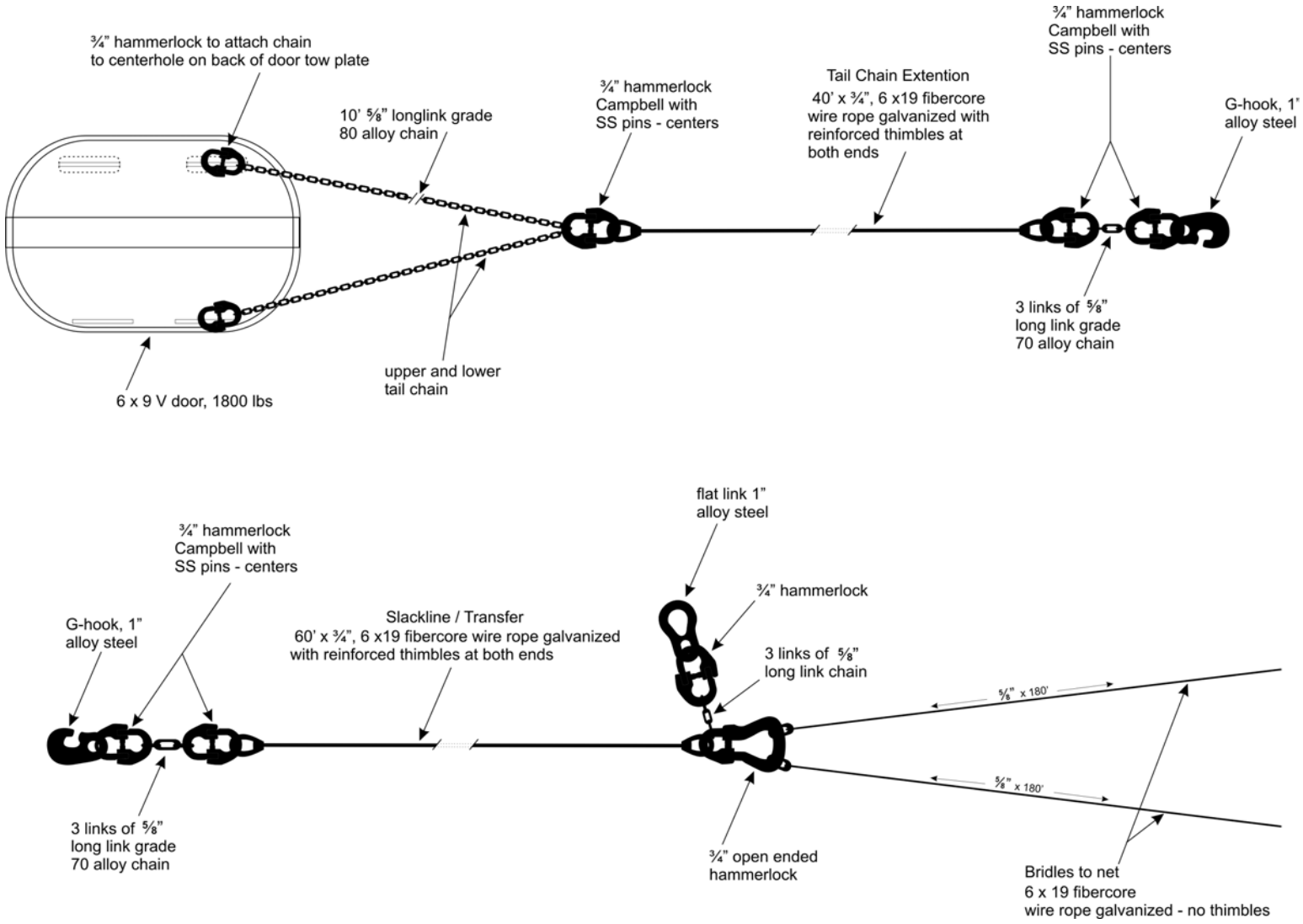
Eastern Bering Sea Shelf Bottom Trawl Survey
Net Reel/Bridle Rigging Plan for 83-112 Eastern Trawl



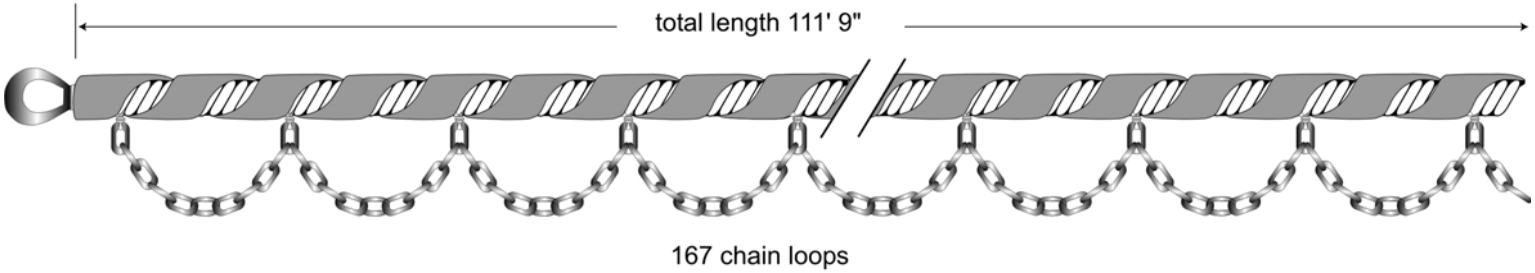
Eastern Bering Sea Shelf Bottom Trawl Survey
Trawl Door Rigging Plan Detail for 83-112 Eastern Trawl
Sole Manufacturer NET Systems, Inc., Bainbridge Island, WA

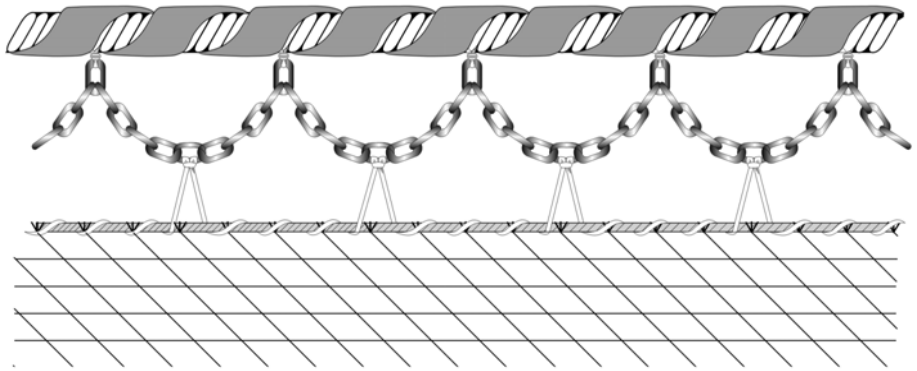


Eastern Bering Sea Shelf Bottom Trawl Survey
Trawl Door Rigging Plan for 83-112 Eastern Trawl



Eastern Bering Sea Shelf Bottom Trawl Survey
Footrope Construction Plan for 83-112 Eastern Trawl





Vessel	
Cruise	
Date	
Checked by	
Checked by	
Net I.D. number	

Survey Trawl Check List
Eastern Bering Sea Shelf Bottom Trawl Survey
83-112 Eastern Trawl

